北京郵電大學

Queen Mary University of London

Joint Programme Assessments 2022/23

Answer Sheet

(To be used in conjunction with the provided alternative assessment questions and the supplementary instructions)

Complete the information below about yourself very carefully.

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Class number	2020215107
Module code	EBU 6018
Word Count	436

I hereby confirm that the submitted work is entirely my own and I have not (i) used the services of any agency or person(s) providing specimen, model or ghostwritten work in the preparation of the work I submit for this open book assessment; (ii) given assistance in accessing this paper or in providing specimen, model or ghostwritten work to other candidates submitting for this open-book assessment.

Signature *Tangxun Wu*

For examiners ONLY

Do not write in this section

	Q1	Q2	Q3	Q4			Sum
A							
B							
C							
D							
E							
F							
TOTAL							

Question	1	Do not write in this column
Section(s)		

x = [-0.98, 1.11, -3.21, -1.31]

y = [1.27, 0.03, -2.39, 5.01]

so that the X:

$$X = \begin{bmatrix} -0.98 & 1.11 & -3.21 & -1.31 \\ 1.27 & 0.03 & -2.39 & 5.01 \end{bmatrix}$$

therefore, through the equation of mean vector:

$$E(X) = \frac{1}{N} \sum_{i=0}^{N-1} \vec{x_i}$$

we can get E(x) = -1.0975, E(y) = 0.9800

so the mean vector of X:

$$E(X) = \begin{bmatrix} -1.0975 \\ 0.9800 \end{bmatrix}$$

Then, we should find the covariance matrix.

$$\begin{aligned} Cov_{x,y} &= \frac{1}{N-1} \sum_{0}^{N-1} (x - x_{mean})(y - y_{mean}) \\ &= \frac{1}{3} \sum_{0}^{3} [(-0.98 + 1.0975)(1.27 - 0.98) + (1.11 + 1.0975)(0.03 - 0.98) \\ &\quad + (-3.21 + 1.0975)(-2.39 - 0.98) + (-1.31 + 1.0975)(5.01 - 0.98)] \\ &= 1.3999 \end{aligned}$$

$$\begin{aligned} Var(x) &= \frac{1}{N-1} \sum_{0}^{N-1} (x_i - x_{mean})^2 \\ &= \frac{1}{3} \sum_{0}^{3} [(-0.98 + 1.0975)^2 + (1.11 + 1.0975)^2 + (-3.21 + 1.0975)^2 + \\ &\quad (-1.31 + 1.0975)^2] \\ &= 3.1316 \end{aligned}$$

Question	1	Do not write in this column
Section(s)		

$$Var(y) = \frac{1}{N-1}\sum_{0}^{N}(y_i - y_{mean})^2$$

$$= \frac{1}{3}\sum_{0}^{3}[(1.27-0.98)^2 + (0.103-0.98)^2 + (-2.39-0.98)^2 + (5.01-0.98)^2]$$

$$= 9.5281$$

Through the values I get and the process of calculation,

I can get the following covariance matrix:

$$\begin{bmatrix} Var(x) & Cov(xy) \\ Cov(xy) & Var(y) \end{bmatrix} = \begin{bmatrix} 3.1316 & 1.3999 \\ 1.3999 & 9.5281 \end{bmatrix}$$

Over.

Question	2	Do not write in this column
Section(s)	i)	

From the question,

s[n] = [1.9, 1.2, 0.9, -0.9, -0.7, -0.8, 0.1, 1.6]

and we know that one expression of DCT[k] is:

$$DCT[k] = c(k)\sum_{n=0}^{N-1} s[n]\cos\frac{\pi(2n+1)k}{2N},\quad k=0,1,2,\cdots,N-1$$

$$c(k)=\begin{cases}\sqrt{\frac{1}{N}}, & k=0\\ \sqrt{\frac{2}{N}}, & k\neq 0\end{cases}$$ c(k) is the normalized factor

another expression of DCT[k] is:

$$DCT[k] = \langle s, \psi_k\rangle$$

$$\psi_k = c(k)\sum_{n=0}^{N-1}\cos\frac{\pi(2n+1)k}{2N},\quad k=0,1,\cdots,N-1$$

and from the book, we know that 8*8 DCT matrix is:

8x8 DCT Matrix

$$\Psi = \frac{1}{2}\begin{bmatrix}
.71 & .71 & .71 & .71 & .71 & .71 & .71 & .71\\
.98 & .83 & .56 & .19 & -.19 & -.56 & -.83 & -.98\\
.92 & .38 & -.38 & -.92 & -.92 & -.38 & .38 & .92\\
.83 & -.19 & -.98 & -.56 & .56 & .98 & .19 & -.83\\
.71 & -.71 & -.71 & .71 & .71 & -.71 & -.71 & .71\\
.56 & -.98 & .19 & .83 & -.83 & -.19 & .98 & -.56\\
.38 & -.92 & .92 & -.38 & -.38 & .92 & -.92 & .38\\
.19 & -.56 & .83 & -.98 & .98 & -.83 & .56 & -.19
\end{bmatrix}$$

so the expression of DCT output sequence y[n]:

$$y[n] = \langle s[n], \psi_8\rangle$$

and the result is:

y[n] = [1.1667, 1.0572, 2.5855, -0.7607, 0.1768, -0.3734, 0.4215, 0.5285]

Question	2	Do not write in this column
Section(s)	ii)	

y[n] = [1.1667, 1.0572, 2.5855, -0.7607, 0.1768, -0.3734, 0.4215, 0.5285]

1. we know that the first output can tell us about the mean value.

So, the mean value of the sequence is 1.1667 divided by sqrt(N)

We get the result: mean value = 0.41249

2. from the output y[n], the third element is very big, this shows that there is a high correlation between the input data and the third row of the transform matrix.

3. the input data has a relatively large variation, and it has a relatively fast rate of change in values, which can be proved by the change from positive value to negative value in the sequence s[n].

